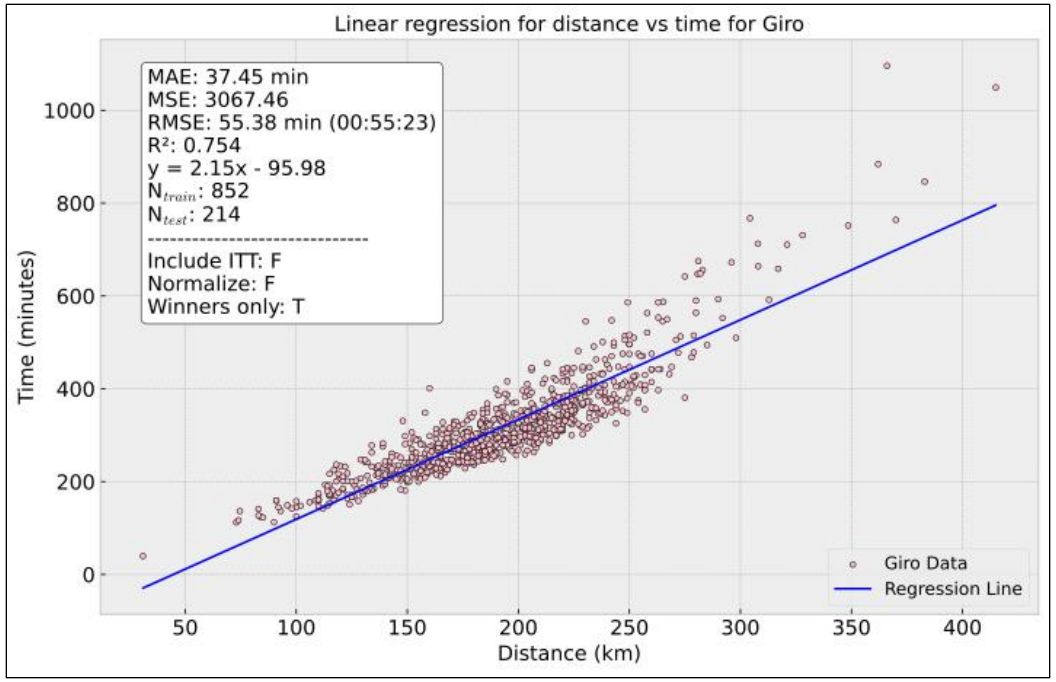
Grand-Tour-Analysis (GTA) group report for week of 3/18/25

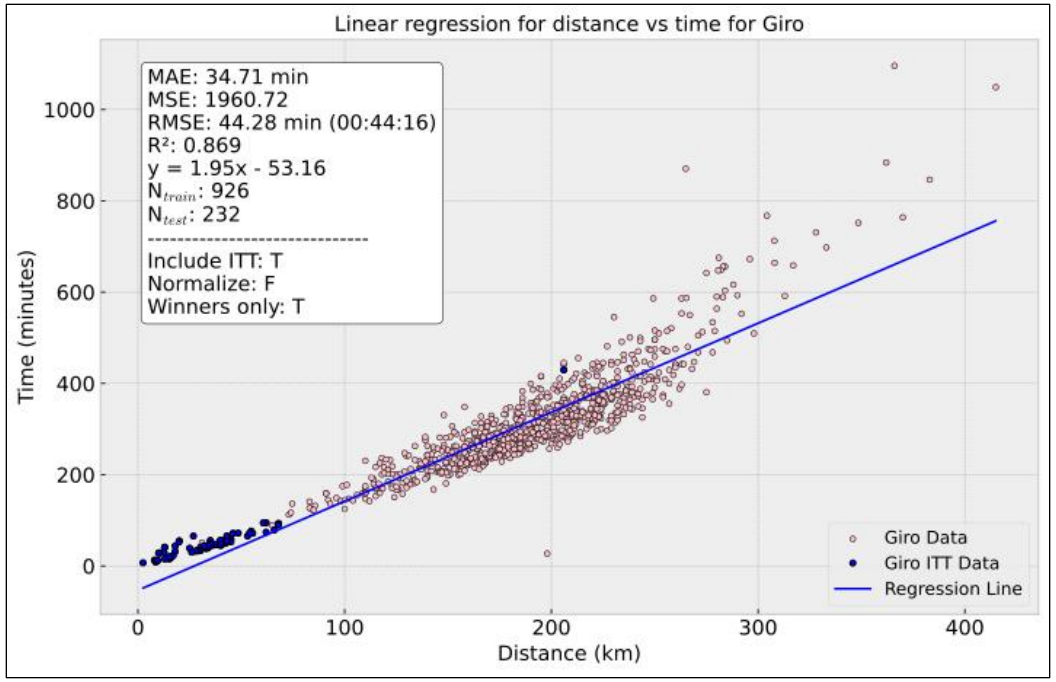
- **INTRODUCTION**: the cycling Grand Tours are Tour de France (TDF) , Giro d'Italia (GRO), and Vuelta a España (LAV). The modern format of each of these races involves cycling 3000+ km around the respective countries. The races are competed by 20 to 22 teams of 8 to 9 riders each in three weeks of racing with extra two days of rest between the racing days. The daily courses/routes of the race-days are called **stages.** Every stage is a timed competition and each day there is an individual winner. On some of the shorter stages, the riders compete for shortest time in what's called Individual Time Trials **(ITT).**

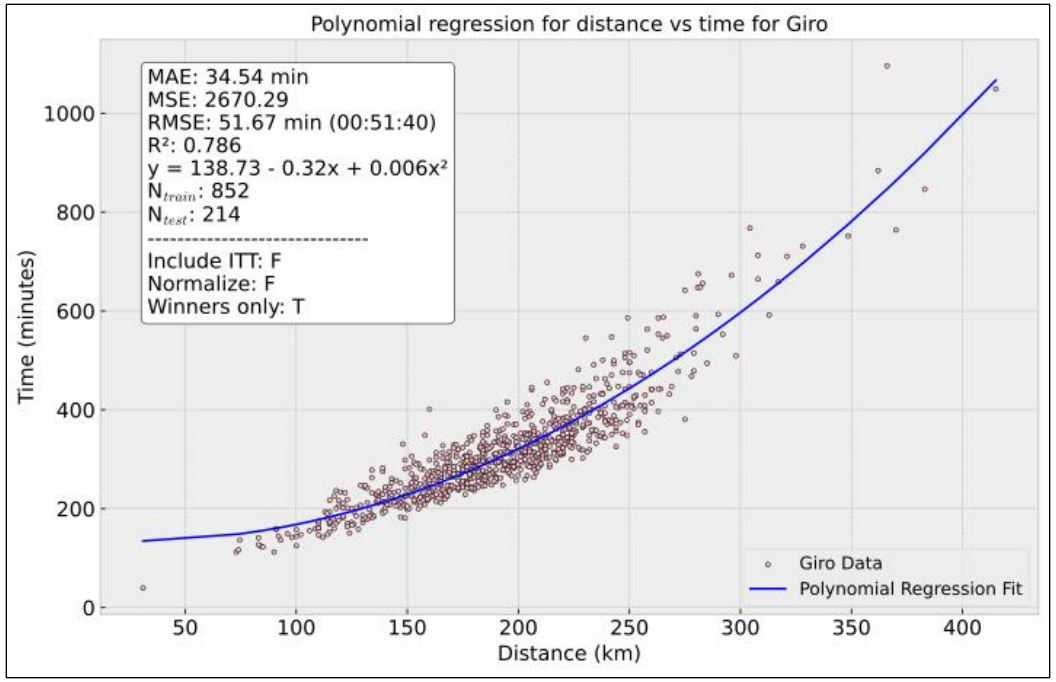
- **DATA**: one of our two databases is called grand_tours which contains all the historic information about the tours. The database (scraped from https://www.procyclingstats.com), among other things, contains information about the year, distance, time, winning time, and speed of all competing riders for every stage of each tour.

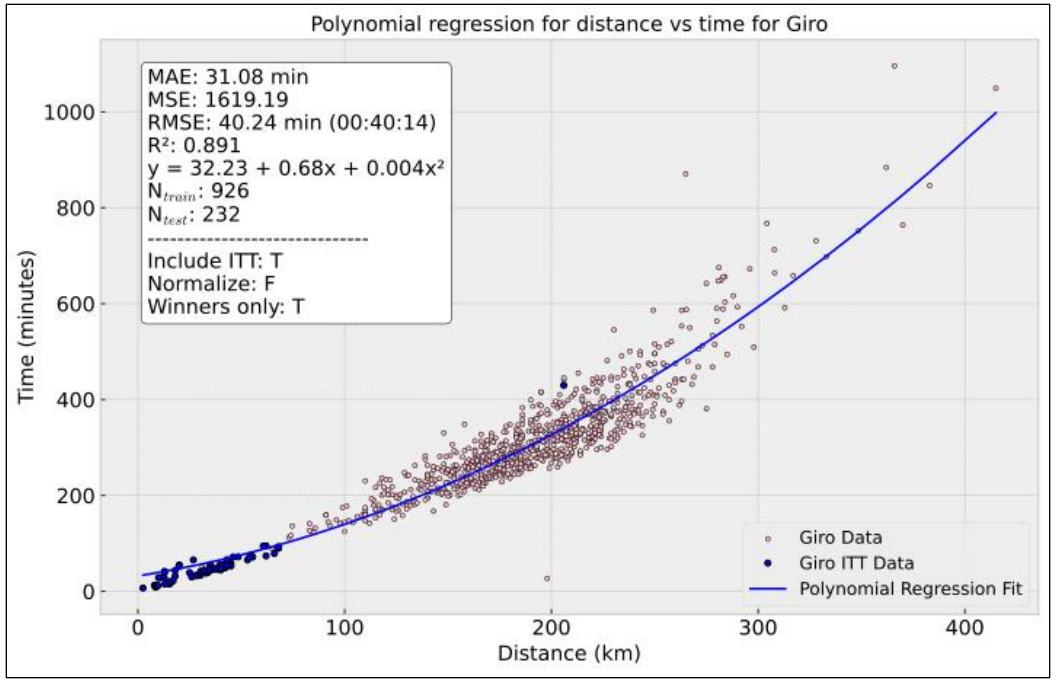
- **WEEKLY GOAL:** get a baseline model
  - From stage-distance vs time data, using a linear regression type model to predict **winning time**.
  - From stage-distance vs speed data, use a linear regression type model to predict **winning speed** (not shown below).

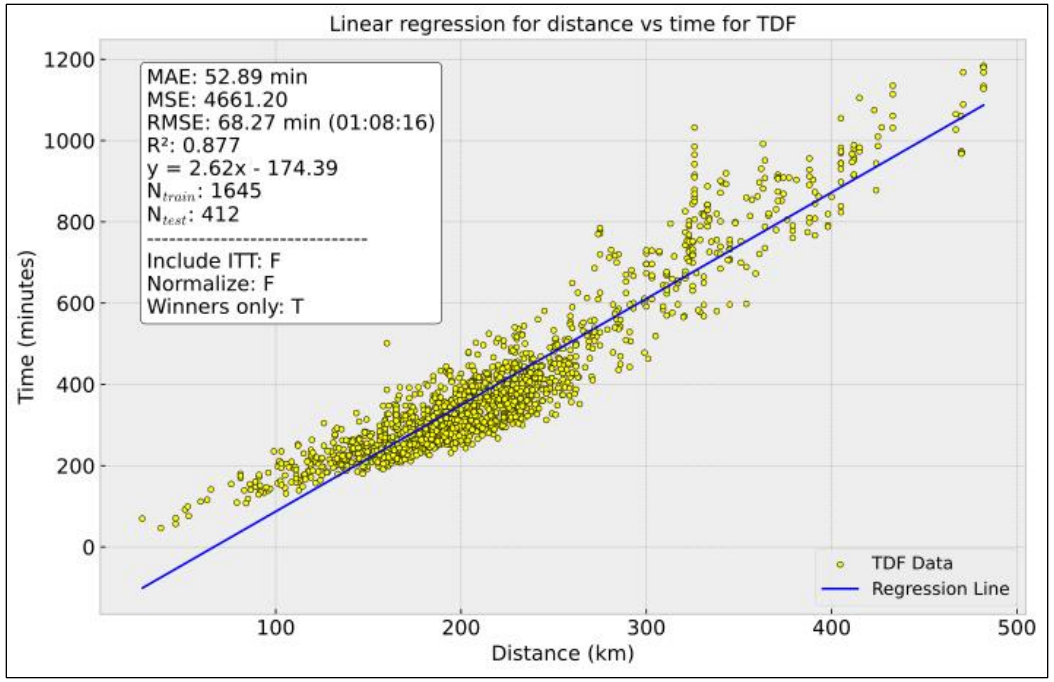
- **RESULTS**: we used sklearn to do linear and second-degree polynomials fit (see plots below) with 80/20 train-test divide. We further broke up the training by whether we
  - include winning times and winning speeds only (only winning times are included so far)
  - include individual time trials or not, and
  - normalize the data or not: stage-distances which range from 0 – 400+ km are mapped to ~(-4,4).

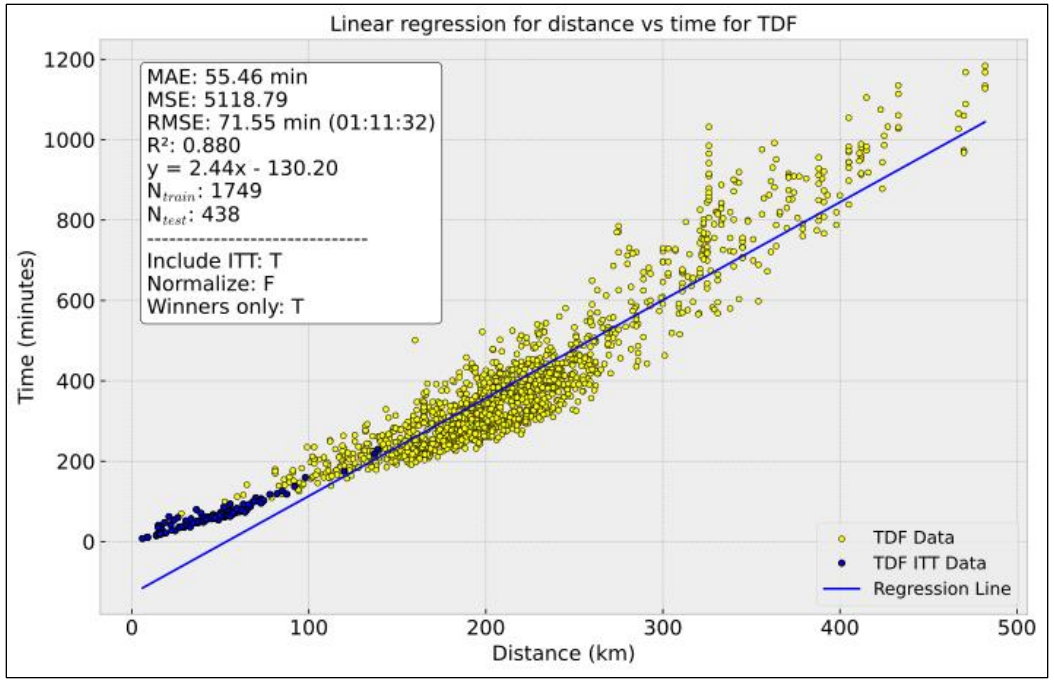

Linear regression for distance vs time for Giro
MAE: 37.45 min
MSE: 3067.46
RMSE: 55.38 min (00:55:23)
R²: 0.754
y = 2.15x - 95.98
N_train: 852
N_test: 214
Include ITT: F
Normalize: F
Winners only: T
Giro Data
Regression Line
Time (minutes)
Distance (km)

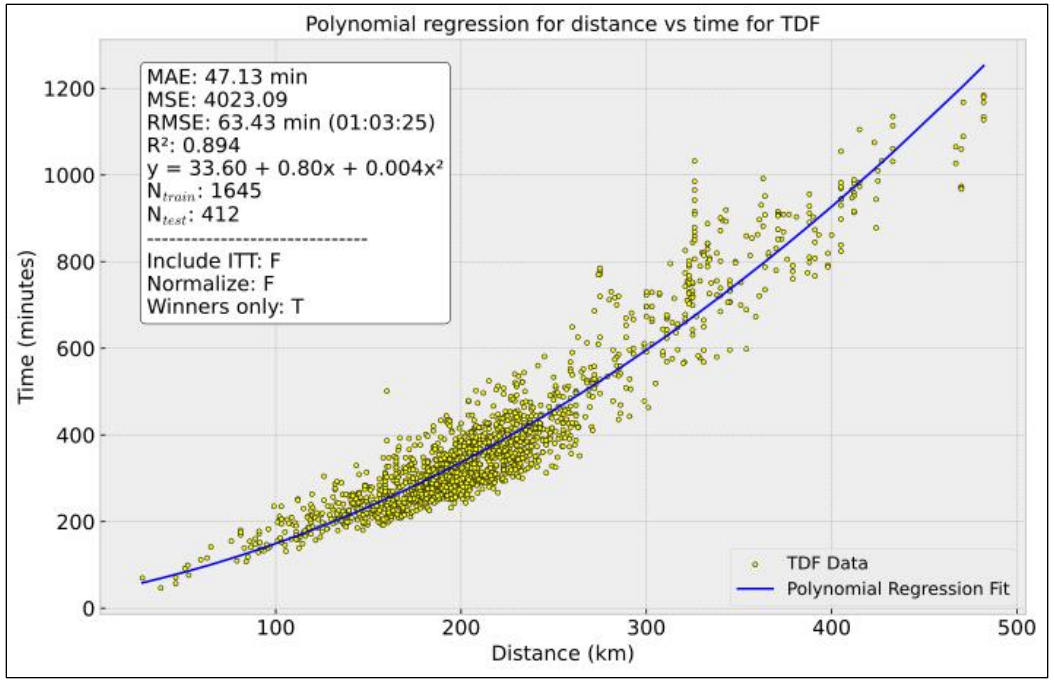

Linear regression for distance vs time for Giro
MAE: 34.71 min
MSE: 1960.72
RMSE: 44.28 min (00:44:16)
R²: 0.869
y = 1.95x - 53.16
N_train: 926
N_test: 232
Include ITT: T
Normalize: F
Winners only: T
Giro Data
Giro ITT Data
Regression Line
Time (minutes)
Distance (km)

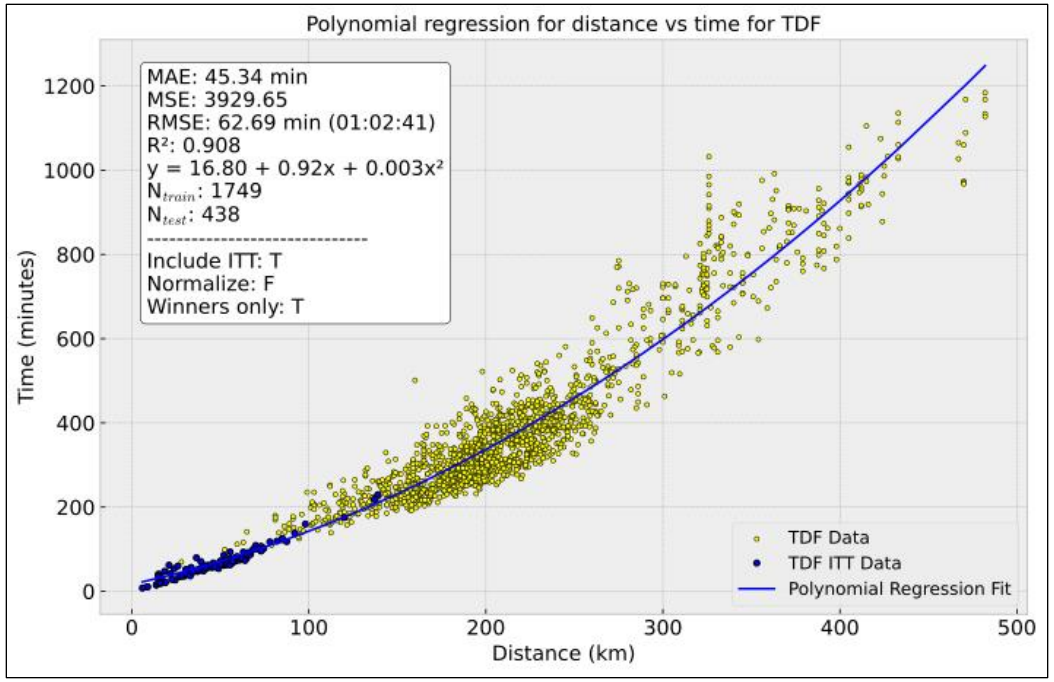

Polynomial regression for distance vs time for Giro
MAE: 34.54 min
MSE: 2670.29
RMSE: 51.67 min (00:51:40)
R²: 0.786
y = 138.73 - 0.32x + 0.006x²
N_train: 852
N_test: 214
Include ITT: F
Normalize: F
Winners only: T
Giro Data
Polynomial Regression Fit
Time (minutes)
Distance (km)

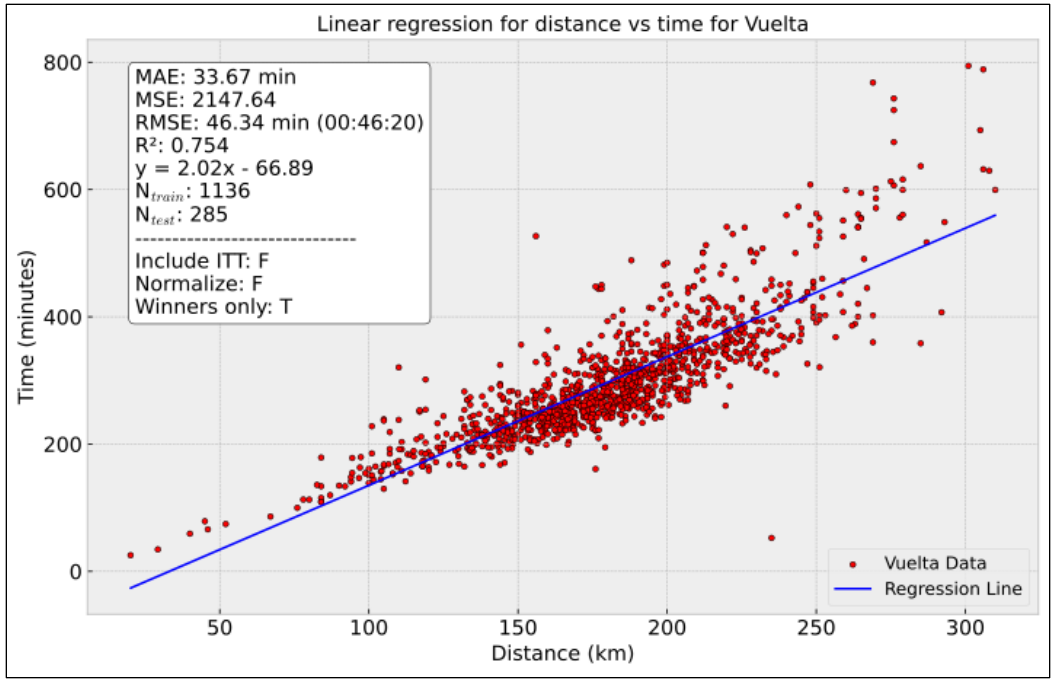

Polynomial regression for distance vs time for Giro
MAE: 31.08 min
MSE: 1619.19
RMSE: 40.24 min (00:40:14)
R²: 0.891
y = 32.23 + 0.68x + 0.004x²
N_train: 926
N_test: 232
Include ITT: T
Normalize: F
Winners only: T
Giro Data
Giro ITT Data
Polynomial Regression Fit
Time (minutes)
Distance (km)

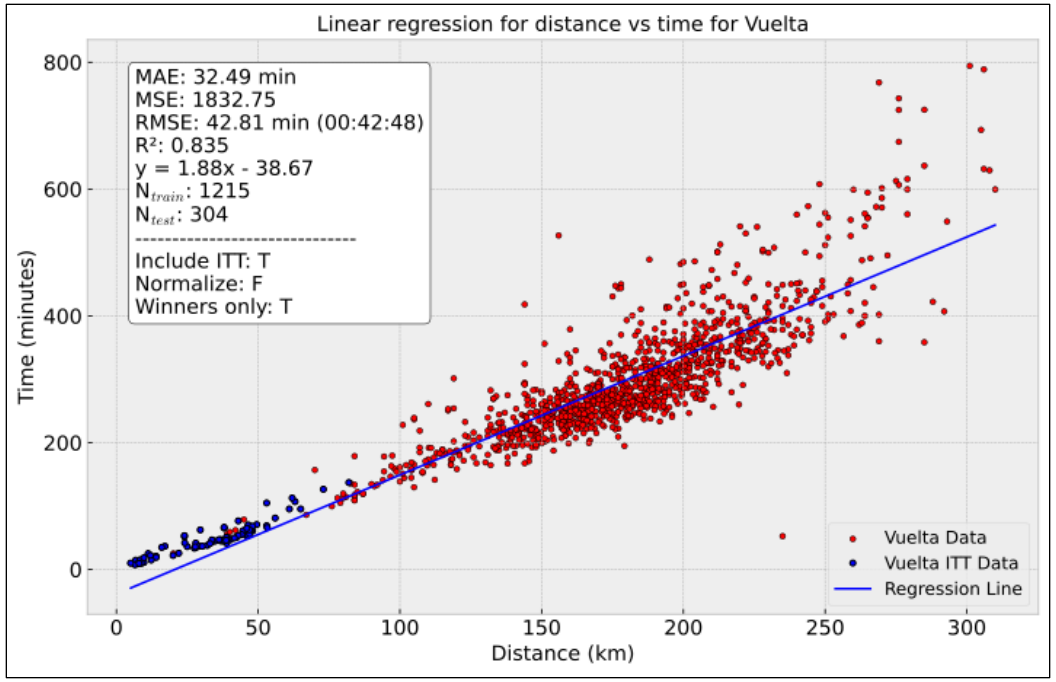

Linear regression for distance vs time for TDF
MAE: 52.89 min
MSE: 4661.20
RMSE: 68.27 min (01:08:16)
R²: 0.877
y = 2.62x - 174.39
N_train: 1645
N_test: 412
Include ITT: F
Normalize: F
Winners only: T
TDF Data
Regression Line
Time (minutes)
Distance (km)

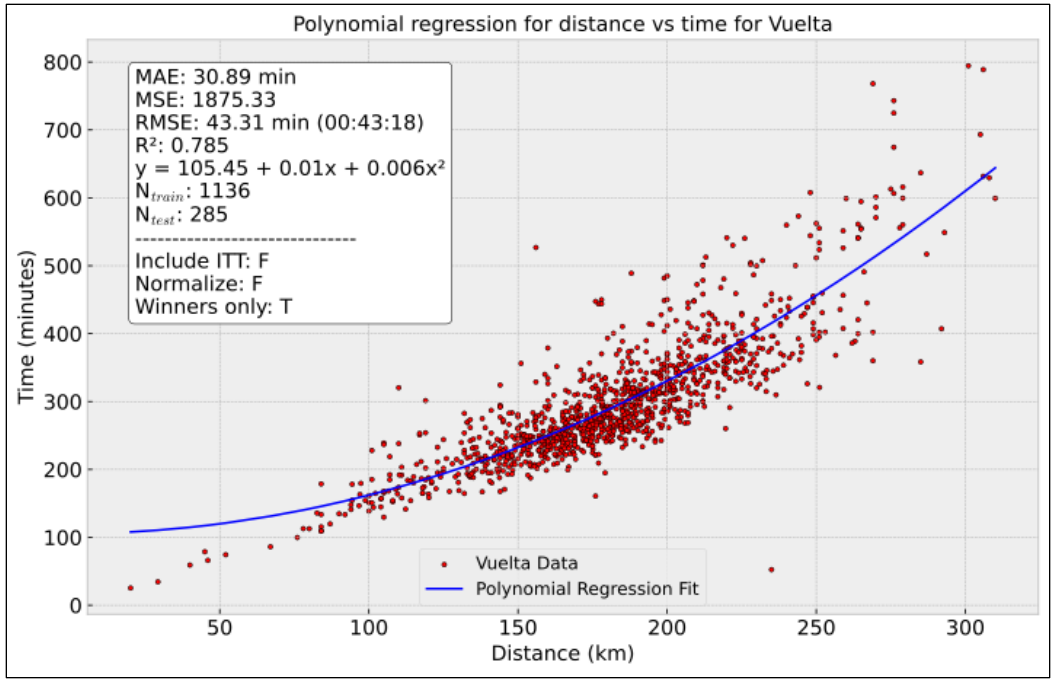

Linear regression for distance vs time for TDF
MAE: 55.46 min
MSE: 5118.79
RMSE: 71.55 min (01:11:32)
R²: 0.880
y = 2.44x - 130.20
N_train: 1749
N_test: 438
Include ITT: T
Normalize: F
Winners only: T
TDF Data
TDF ITT Data
Regression Line
Time (minutes)
Distance (km)

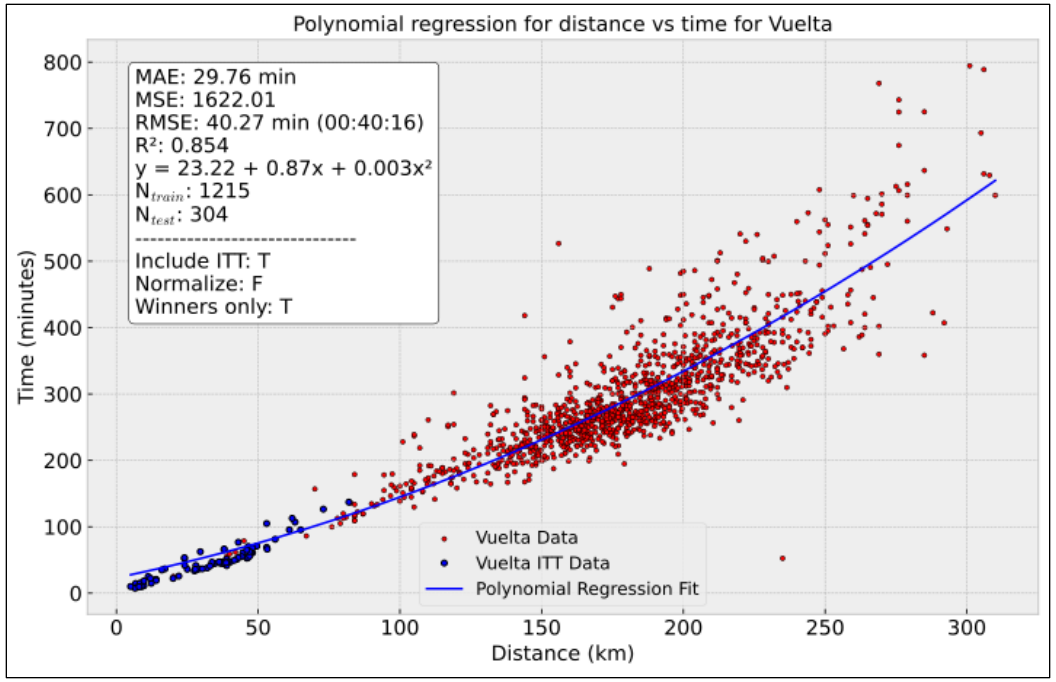

Polynomial regression for distance vs time for TDF
MAE: 47.13 min
MSE: 4023.09
RMSE: 63.43 min (01:03:25)
R²: 0.894
y = 33.60 + 0.80x + 0.004x²
N_train: 1645
N_test: 412
Include ITT: F
Normalize: F
Winners only: T
TDF Data
Polynomial Regression Fit
Time (minutes)
Distance (km)


Polynomial regression for distance vs time for TDF
MAE: 45.34 min
MSE: 3929.65
RMSE: 62.69 min (01:02:41)
R²: 0.908
y = 16.80 + 0.92x + 0.003x²
N_train: 1749
N_test: 438
Include ITT: T
Normalize: F
Winners only: T
TDF Data
TDF ITT Data
Polynomial Regression Fit
Time (minutes)
Distance (km)


Linear regression for distance vs time for Vuelta
MAE: 33.67 min
MSE: 2147.64
RMSE: 46.34 min (00:46:20)
R²: 0.754
y = 2.02x - 66.89
N_train: 1136
N_test: 285
Include ITT: F
Normalize: F
Winners only: T
Vuelta Data
Regression Line
Time (minutes)
Distance (km)


Linear regression for distance vs time for Vuelta
MAE: 32.49 min
MSE: 1832.75
RMSE: 42.81 min (00:42:48)
R²: 0.835
y = 1.88x - 38.67
N_train: 1215
N_test: 304
Include ITT: T
Normalize: F
Winners only: T
Vuelta Data
Vuelta ITT Data
Regression Line
Time (minutes)
Distance (km)


Polynomial regression for distance vs time for Vuelta
MAE: 30.89 min
MSE: 1875.33
RMSE: 43.31 min (00:43:18)
R²: 0.785
y = 105.45 + 0.01x + 0.006x²
N_train: 1136
N_test: 285
Include ITT: F
Normalize: F
Winners only: T
Vuelta Data
Polynomial Regression Fit
Time (minutes)
Distance (km)


Polynomial regression for distance vs time for Vuelta
MAE: 29.76 min
MSE: 1622.01
RMSE: 40.27 min (00:40:16)
R²: 0.854
y = 23.22 + 0.87x + 0.003x²
N_train: 1215
N_test: 304
Include ITT: T
Normalize: F
Winners only: T
Vuelta Data
Vuelta ITT Data
Polynomial Regression Fit
Time (minutes)
Distance (km)

Future goals for first-half of our data (grand_tours.db) only

- Add another feature from the database to do a multi-dimensional regression. Interesting features in the database include
  - Stage number: we expect the times to be higher and speeds to be lower for later stages because the riders will be tired.
  - Vertical distance (m): we expect times the times to be higher and speeds to be lower for higher climbs
  - Average temperature: the times to be higher and speeds to be lower for hotter temperatures
  - Profile score: a custom regressor used by procyclingstats to rank stages
  - Starlist quality (procyclingstats related feature)
- Add a predictor model where given (stage length, vertical distance, …) it tries to predict how the race will be decided (solo ride, sprint, …)